

# SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

## CHAPTER 04

# Carbon and Its Compounds

(Full Chapter)

Covalent Bonding • Versatile Nature of Carbon • Homologous Series •  
Nomenclature & Functional Groups • Chemical Properties of Carbon Compounds •  
Ethanol & Ethanoic Acid • Soaps & Detergents

Target Level: CBSE Class 10

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## Chapter Overview & Basic Concepts

Carbon is a versatile non-metal element present in all living structures, food, clothes, and fuels. The Earth's crust contains only 0.02% carbon in minerals, and the atmosphere has 0.03% CO<sub>2</sub>.

- Carbon always forms **covalent bonds** by sharing valence electrons to attain noble gas configuration.
- Atomic number of Carbon = 6 (Electronic Configuration: 2, 4; Valency = 4).
- It cannot lose 4 electrons to form C<sup>4+</sup> because it requires huge energy, nor can it gain 4 electrons to form C<sup>4-</sup> as 6 protons cannot hold 10 electrons.

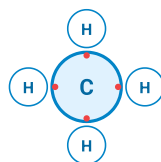
**NCERT High-Yield Insight:** Covalent compounds have strong bonds within the molecule, but weak inter-molecular forces. Consequently, they have low melting and boiling points and are generally poor conductors of electricity (no free ions or free electrons).

### SECTION 1: COVALENT BONDING & VERSATILE NATURE OF CARBON

#### 1.1 Types of Covalent Bonds

- **Single Bond:** Sharing 1 pair of electrons (e.g., H<sub>2</sub>, Cl<sub>2</sub>, CH<sub>4</sub>).
- **Double Bond:** Sharing 2 pairs of electrons (e.g., O<sub>2</sub>, CO<sub>2</sub>).
- **Triple Bond:** Sharing 3 pairs of electrons (e.g., N<sub>2</sub>).

 **Diagram: Electron Dot Structure of Methane (CH<sub>4</sub>)**



#### 1.2 Two Reasons for Versatile Nature of Carbon

- **Catenation:** Unique ability of carbon to form direct covalent bonds with other carbon atoms, forming long straight chains, branched chains, or closed ring structures.
- **Tetravalency:** Having 4 valence electrons, carbon can bond with 4 other monovalent atoms or atoms of oxygen, nitrogen, sulfur, and hydrogen.

## SECTION 2: HYDROCARBONS & HOMOLOGOUS SERIES

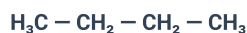
| Feature         | Saturated Hydrocarbons (Alkanes)              | Unsaturated Hydrocarbons (Alkenes & Alkynes)              |
|-----------------|-----------------------------------------------|-----------------------------------------------------------|
| Bonds           | Single bond between Carbon-Carbon (C-C).      | Double (C=C) or Triple (C≡C) bonds between Carbon-Carbon. |
| General Formula | $C_nH_{2n+2}$ (Alkane)                        | $C_nH_{2n}$ (Alkene)   $C_nH_{2n-2}$ (Alkyne)             |
| Reactivity      | Relatively unreactive (undergo substitution). | Highly reactive (undergo addition reactions).             |
| Flame Type      | Burn with a clean blue flame.                 | Burn with a yellow smoky flame (incomplete combustion).   |

### 2.1 Homologous Series & Structural Isomerism

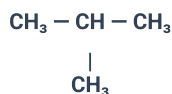
- Homologous Series:** A series of organic compounds having the same functional group and similar chemical properties, where successive members differ by a **-CH<sub>2</sub>- group** and a molecular mass of **14 u**.
- Structural Isomers:** Compounds having the same molecular formula but different structural arrangements (e.g., n-Butane and Iso-butane for C<sub>4</sub>H<sub>10</sub>).

#### Diagram: Structural Isomers of Butane (C<sub>4</sub>H<sub>10</sub>)

n-Butane (Straight Chain)



Iso-butane (Branched)



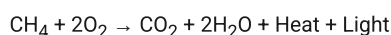
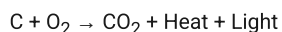
### 2.2 Functional Groups Overview

| Heteroatom | Functional Group      | Formula  | Example Compound (IUPAC Name)                  |
|------------|-----------------------|----------|------------------------------------------------|
| Cl / Br    | Halo (Chloro / Bromo) | -Cl, -Br | Chloromethane (CH <sub>3</sub> Cl)             |
| Oxygen     | Alcohol               | -OH      | Ethanol (C <sub>2</sub> H <sub>5</sub> OH)     |
| Oxygen     | Aldehyde              | -CHO     | Ethanal (CH <sub>3</sub> CHO)                  |
| Oxygen     | Ketone                | >C=O     | Propanone (CH <sub>3</sub> COCH <sub>3</sub> ) |
| Oxygen     | Carboxylic Acid       | -COOH    | Ethanoic Acid (CH <sub>3</sub> COOH)           |

## Science Master X Class 10 Science (Chapter 4)

### SECTION 3: CHEMICAL PROPERTIES OF CARBON COMPOUNDS

**1. Combustion:** Carbon compounds burn in oxygen to produce  $\text{CO}_2$ ,  $\text{H}_2\text{O}$ , heat, and light.



**2. Oxidation:** Ethanol is oxidized to ethanoic acid using alkaline  $\text{KMnO}_4$  or acidified  $\text{K}_2\text{Cr}_2\text{O}_7$ .



**3. Addition Reaction:** Unsaturated hydrocarbons add hydrogen in presence of Nickel (Ni) / Palladium (Pd) catalyst to give saturated hydrocarbons (Vegetable oils  $\rightarrow$  Vegetable ghee).



**4. Substitution Reaction:** Saturated alkanes react with Chlorine in presence of sunlight.



### SECTION 4: IMPORTANT CARBON COMPOUNDS & SOAPS

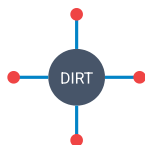
#### 4.1 Ethanol vs. Ethanoic Acid

| Property                               | Ethanol ( $\text{C}_2\text{H}_5\text{OH}$ )                                                                                                                                                     | Ethanoic Acid ( $\text{CH}_3\text{COOH}$ )     |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Common Name                            | Ethyl Alcohol                                                                                                                                                                                   | Acetic Acid (5-8% solution in water = Vinegar) |
| Litmus Test                            | Neutral (No color change)                                                                                                                                                                       | Turns blue litmus red                          |
| Esterification Reaction                | $\text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH} \rightarrow (\text{Conc. H}_2\text{SO}_4) \rightarrow \text{CH}_3\text{COOC}_2\text{H}_5$ (Sweet smelling Ester) + $\text{H}_2\text{O}$ |                                                |
| Saponification                         | Ester + $\text{NaOH} \rightarrow \text{CH}_3\text{COONa}$ (Sodium Acetate / Soap) + $\text{C}_2\text{H}_5\text{OH}$                                                                             |                                                |
| Reaction with $\text{Na}_2\text{CO}_3$ | No reaction                                                                                                                                                                                     | Gives effervescence of $\text{CO}_2$ gas       |

#### 4.2 Soaps and Detergents (Cleansing Action)

- Soap Structure:** Sodium or potassium salt of long-chain fatty acids.
  - Ionic End (Head):** Hydrophilic (water-loving), faces outside towards water.
  - Hydrocarbon Chain (Tail):** Hydrophobic (water-fearing), attaches to dirt/oil.
- Micelle Formation:** Spherical aggregate formed by soap molecules in water trapped around oily dirt particle.
- Hard Water Issue:** Soaps form insoluble precipitate called **scum** with  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  ions in hard water. Detergents (ammonium/sulfonate salts) do not form scum.

 **Diagram: Structure of Soap Micelle**



**Science Master X Class 10 Science (Chapter 4)**

**? Important Practice Questions**

**Q1: What are covalent bonds? Why do covalent compounds have low melting points and poor electrical conductivity?**

**Ans:** Covalent bonds are formed by sharing of electron pairs between atoms. They have low melting points because intermolecular forces are weak, and poor conductivity because no free electrons or charged ions are formed.

**Q2: Define catenation and tetravalency. What is their significance for carbon?**

**Ans:** Catenation is carbon's ability to form long direct chains/rings with other carbon atoms. Tetravalency means carbon has 4 valence electrons to bond with 4 monovalent atoms. Together, they explain the existence of millions of carbon compounds.

**Q3: Differentiate between Saturated and Unsaturated hydrocarbons.**

**Ans:** Saturated hydrocarbons (Alkanes) contain single C-C bonds, burn with a blue clean flame, and are unreactive. Unsaturated hydrocarbons (Alkenes/Alkynes) contain double/triple bonds, burn with a yellow smoky flame, and undergo addition reactions.

**Q4: What is Esterification? Write a balanced chemical equation.**

**Ans:** The reaction between a carboxylic acid and an alcohol in presence of concentrated  $\text{H}_2\text{SO}_4$  to form a sweet-smelling ester.



**Q5: Why do soaps fail to show effective cleansing action in hard water? How do detergents overcome this?**

**Ans:** Hard water contains  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  ions which react with soap to form insoluble sticky precipitate (scum). Detergents do not form scum with hard water because their charged ends do not bind with  $\text{Ca}^{2+}/\text{Mg}^{2+}$  ions.

